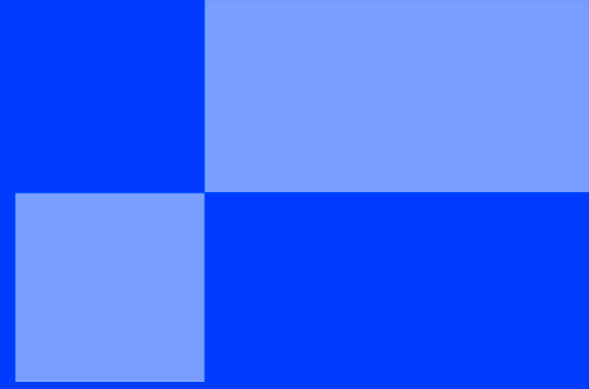




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< BOOTSTRAP INSTRUCTIONS />

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DQN on CartPole-v1 — Quick Start

Setup

```
# Create virtual environment with uv
uv venv .venv --python 3.10
source .venv/bin/activate

# Install dependencies
uv pip install gymnasium torch tensorboard
```

Training

Run the training loop:

```
python3 train_cartpole.py
```

This trains a DQN agent on CartPole-v1 for 500 episodes. You should see the episode reward climb to ~500+ (the max) within 200-500 episodes, indicating the pole has been balanced consistently.

Hyperparameters (edit `train_cartpole.py` if needed)

	Param	Value	Description
GAMMA	0.99	Discount factor for Bellman target	
LR	1e-3	Adam learning rate	
BATCH_SIZE	64	Minibatch size for experience replay	
EPS_START	1.0	Starting epsilon (exploration)	
EPS_END	0.05	Final epsilon	
EPS_DECAY	0.995	Per-episode epsilon decay	
WARMUP	1000	No training until buffer exceeds this	
TARGET_UPDATE	10	Hard update target network every N episodes	

Note: `CUDA_VISIBLE_DEVICES=""` forces CPU-only mode. Remove it if you have a GPU with enough VRAM, or set `CUDA_VISIBLE_DEVICES="0"` to use the first GPU.

Viewing TensorBoard Logs

```
tensorboard --logdir runs/cartpole_dqn
```

Then open `http://localhost:6006` in your browser. You'll see:

- ✓ **Reward/episode_reward** — per-episode total reward (also called "reward per episode")
- ✓ **Reward/reward_per_step** — average reward per time step in the episode
- ✓ **Reward/avg_reward_100** — 100-episode moving average
- ✓ **Length/episode_length** — episodes last (max = 500)
- ✓ **Epsilon/epsilon** — exploration rate over time

Evaluation

Run the agent for 5 episodes with rendering:

```
python3 eval.py
```

This loads `cartpole_dqn.pt`, runs 5 untrained episodes with `render_mode='human'`, and prints the average reward.

v1

{EPITECH}